Representing inputs as vectors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable input-schema and encoding record.

Trace conversions,repair missing inputs and explain information a vector cannot retain.

Keep one habit: W1–W3/W5 → selected battery schema → project input contract → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A neural network receives numbers in a specified order. Before learning how it computes, we need to represent one example without losing the meaning of its fields.

  A **vector** is an ordered list: [4, 2] can mean width 4 cm and height 2 cm if that is the declared order. Reversing the two numbers describes a different rectangle under the same **schema**.

- For this fictional parcel investigation, use [width in cm, height in cm]. Width and height must both be present and greater than zero. The **vector** has **dimension** 2 because it contains two **component**s.

  Record IDs identify examples; they do not measure the parcel. The target class is stored separately.

- There are three examples, each with two features. Stacking their **vector**s makes a table with three rows and two columns. Do not confuse the number of rows with the **dimension** of one **input**.

ML Builder Expo

> **My remembered idea, example and correction**
>
> | |
> |---|
> | |
> | |
> | |

Lesson reference for workbook practice

Record	Width and height in cm	Input vector
A	4 and 2	[4, 2]
B	2 and 4	[2, 4]
C	3 and 3	[3, 3]

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Parcel schema[width_cm, height_cm], both present and positive. Awidth4 height2→[4, 2], Bwidth2 height4→[2, 4], Cwidth3 height3→[3, 3]:dimension 2, stacked table 3 rows×2 columns. Dwidth50 mmheight30 mm→[5, 3]; Ewidth20 mmheight70 mm→[2, 7]. New W2 width60 mmheight10 mm→[6, 1]. Missing height is unknown, not 0. IDs/future target excluded.

  Same[4, 2]can hide different material/labels; a deterministic fixed model using only those coordinates cannot distinguish that pair. Separate brightnessA[2, 7]/B[7, 2]bothsum 9 but preserve different positions. Independent battery task supplies its own schema and age-zero rule.

- Write A, B and C as ordered pairs from the lesson. State the **dimension** of each and the shape of the stacked table.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Parcel schema[width_cm, height_cm], both present and positive. Awidth4 height2→[4, 2], Bwidth2 height4→[2, 4], Cwidth3 height3→[3, 3]:dimension 2, stacked table 3 rows×2 columns. Dwidth50 mmheight30 mm→[5, 3]; Ewidth20 mmheight70 mm→[2, 7]. New W2 width60 mmheight10 mm→[6, 1]. Missing height is unknown, not 0. IDs/future target excluded.

  Same[4, 2]can hide different material/labels; a deterministic fixed model using only those coordinates cannot distinguish that pair. Separate brightnessA[2, 7]/B[7, 2]bothsum 9 but preserve different positions. Independent battery task supplies its own schema and age-zero rule.

- A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Parcel schema[width_cm, height_cm], both present and positive. Awidth4 height2→[4, 2], Bwidth2 height4→[2, 4], Cwidth3 height3→[3, 3]:dimension 2, stacked table 3 rows×2 columns. Dwidth50 mmheight30 mm→[5, 3]; Ewidth20 mmheight70 mm→[2, 7]. New W2 width60 mmheight10 mm→[6, 1]. Missing height is unknown, not 0. IDs/future target excluded.

  Same[4, 2]can hide different material/labels; a deterministic fixed model using only those coordinates cannot distinguish that pair. Separate brightnessA[2, 7]/B[7, 2]bothsum 9 but preserve different positions. Independent battery task supplies its own schema and age-zero rule.

- Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this **schema**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Parcel schema[width_cm, height_cm], both present and positive. Awidth4 height2→[4, 2], Bwidth2 height4→[2, 4], Cwidth3 height3→[3, 3]:dimension 2, stacked table 3 rows×2 columns. Dwidth50 mmheight30 mm→[5, 3]; Ewidth20 mmheight70 mm→[2, 7]. New W2 width60 mmheight10 mm→[6, 1]. Missing height is unknown, not 0. IDs/future target excluded.

  Same[4, 2]can hide different material/labels; a deterministic fixed model using only those coordinates cannot distinguish that pair. Separate brightnessA[2, 7]/B[7, 2]bothsum 9 but preserve different positions. Independent battery task supplies its own schema and age-zero rule.

- Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two **input**s apart.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

  Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new schema independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled representing **input**s as **vector**s **evidence** sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Convert before encoding for Representing **input**s as **vector**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Representing **input**s as **vector**s. Proposed explanation: "Order does not matter". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

  Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Encode the same pixel measurements in a fixed **vector** order.

- Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/1

Representing inputs as vectors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare coordinate order

Coordinate 0 is left brightness; coordinate 1 is right brightness.

2. Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.

3. Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

CHANGE ONE THING

Swap only A's vector order before prediction.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Encode the same pixel measurements in a fixed **vector** order.
- Which labelled step matches this detail? Use the same order when **training** and predicting; record missing measurements rather than guessing.
- 1. Declare coordinate order
- 2. Check **input** contract
- 3. Encode both images
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check **input** contract Use the same order when **training** and predicting; record missing measurements rather than guessing.

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Changed-case question and optional hint

- Changed-case practice: Swap only A's **vector** order before **prediction**. Which conclusion is supported by the supplied facts?
- 1. The encoded **input** becomes [7, 2], numerically identical to B's original **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Repair and retain the record

Review ML Builder Expo (ai-4/36). Goal: Defend the project using traceable **evidence**. Short practice: Which three supporting items should be visible at the expo?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Representing inputs as vectors: the case

AI ACADEMY / AI-5/1 · 01 · LOOK AT THE FACTS

Representing inputs as vectors: the case

The situation

Encode the same pixel measurements in a fixed vector order.

Supplied evidence

Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2. All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain representing inputs as vectors using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

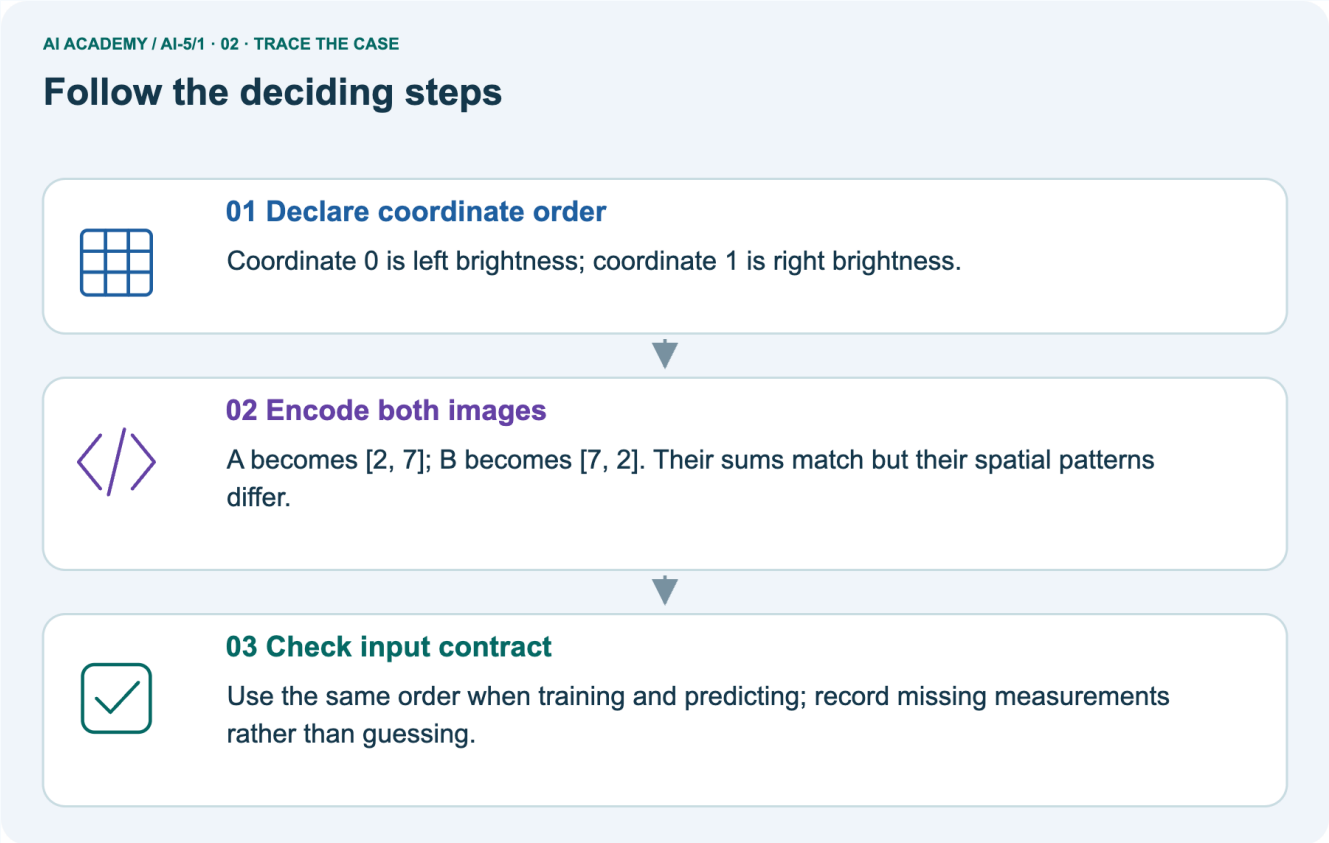
Encode the same pixel measurements in a fixed **vector** order. Supplied facts: Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

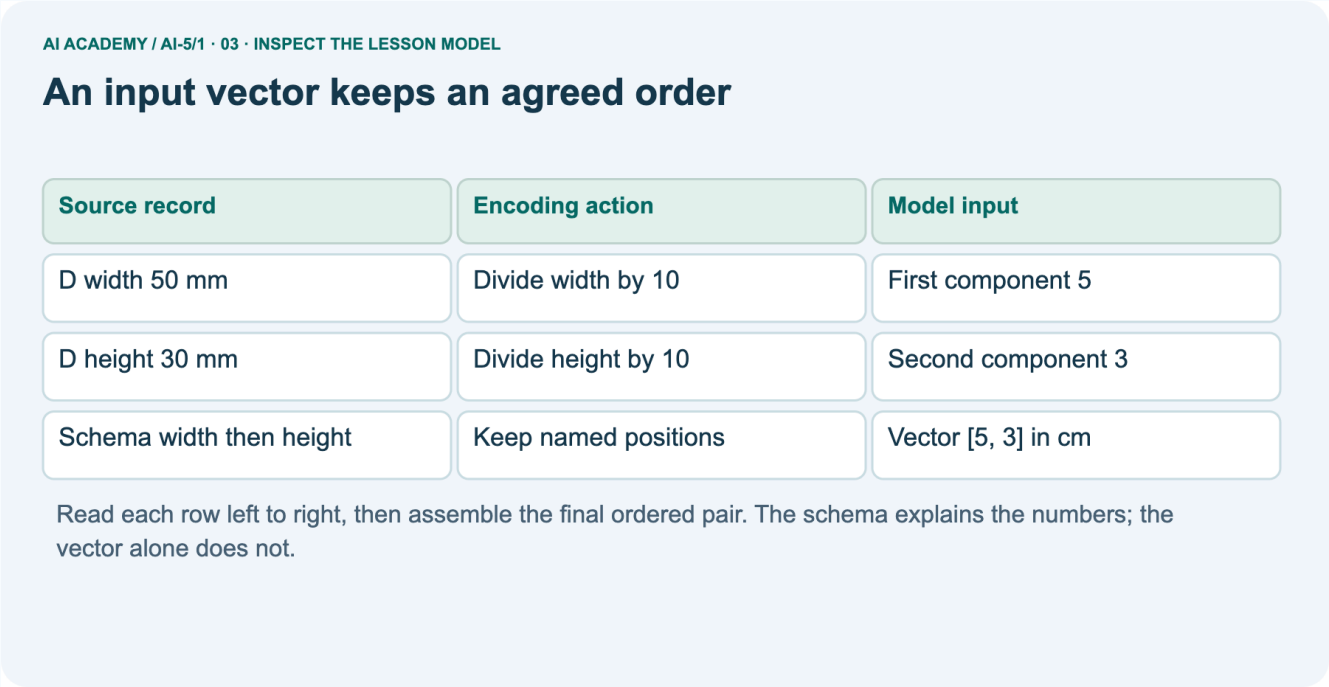
Declare coordinate order: Coordinate 0 is left brightness; coordinate 1 is right brightness. → Encode both images: A becomes [2, 7]; B becomes [7, 2].

Their sums match but their spatial patterns differ. → Check **input** contract: Use the same order when **training** and predicting; record missing measurements rather than guessing.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. An input vector keeps an agreed order



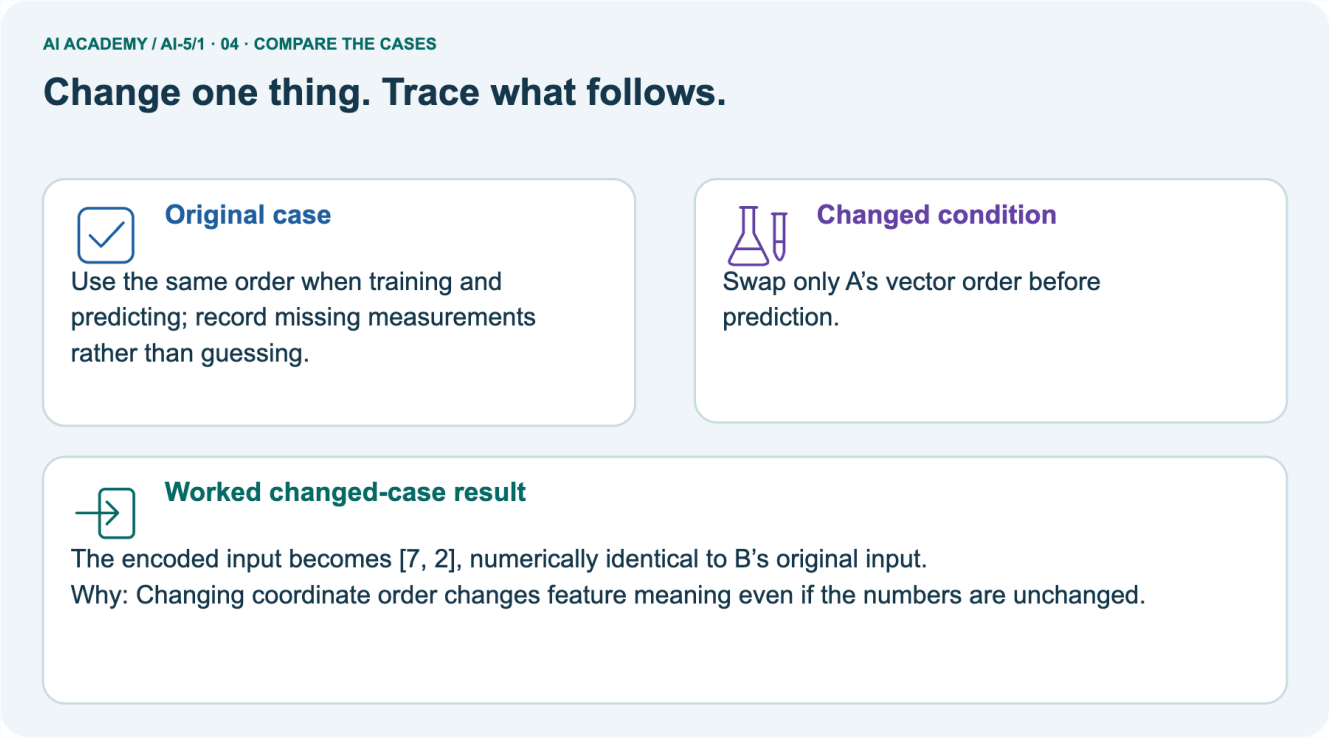
Follow the labelled picture with the matching explanation. Open the vector diagram

An **input vector** keeps an agreed order. D width 50 mm | Divide width by 10 | First **component** 5. D height 30 mm | Divide height by 10 | Second **component** 3. **Schema** width then height | Keep named positions | **Vector** [5, 3] in cm

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

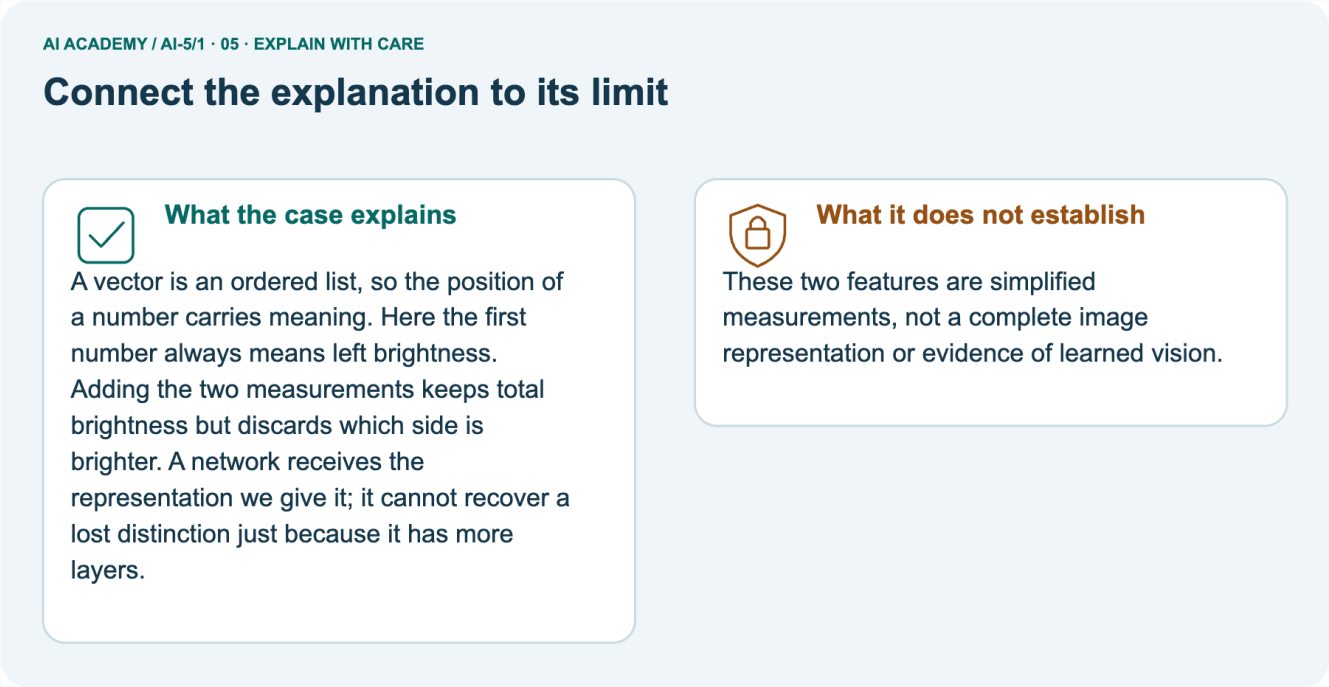
Original case: Use the same order when **training** and predicting; record missing measurements rather than guessing. Changed condition: Swap only A's **vector** order before **prediction**.

Expected result: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

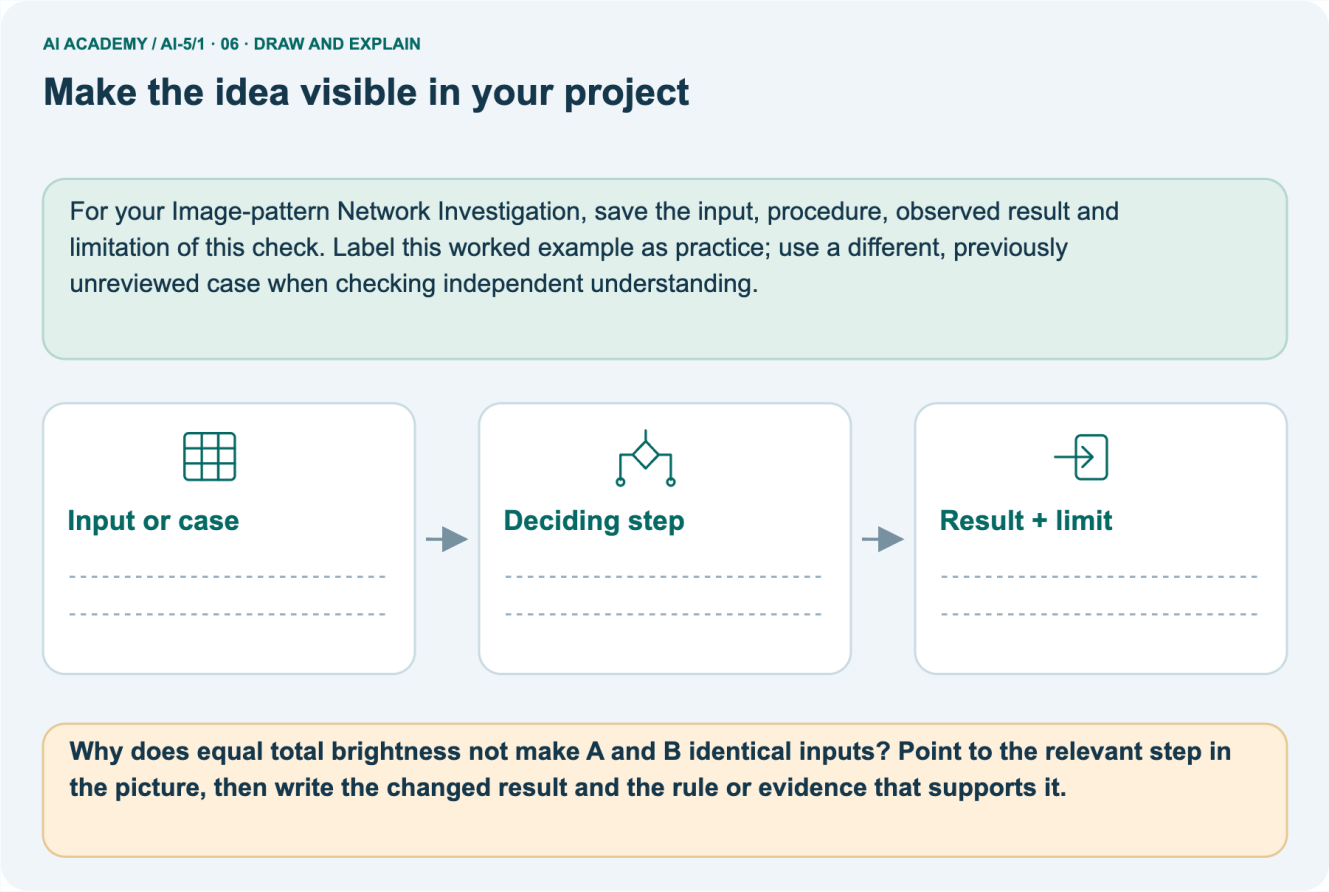
A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Limit: These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Representing inputs as vectors: the case

02. Follow the deciding steps

03. An input vector keeps an agreed order

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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